

VDA9.6

Hugo Lladró Prats

June 2026

Exercise 6 (Finite Differences, 5 Points)

a) Use the Taylor expansion to show that the forward finite difference scheme on a uniform grid with stepsize h is first-order accurate (i.e., the approximation error decreases linearly with decreasing stepsize h). (3P) b) Use the same strategy to show that the central finite difference scheme is second-order accurate (i.e., the approximation error decreases with h^2). (2P)

a)

As both methods compute the values of the directional derivatives by only switching the values in the corresponding component, we can reduce our analysis to one-dimensional functions. We therefore assume that for each component x, y, z (for example x), we can fix the values of y, z to y_0, z_0 , and obtain the function $g(x) := f(x, y_0, z_0)$, which we will assume, is C^2 at the desired point. Therefore, by the Taylor expansion theorem, we would have that $g(x) = g(x_0) + (x - x_0)g'(x_0) + \frac{1}{2}(x - x_0)^2g''(x_0) + o((x - x_0)^2)$. If we set $h = x - x_0$, we have that the expression for the forward finite difference scheme would be $g'(x_0) \approx \frac{1}{h}(g(x_0 + h) - g(x_0)) = \frac{1}{h}(g(x_0) + hg'(x_0) + \frac{1}{2}h^2g''(x_0) + o(h^2) - g(x_0)) = g'(x_0) + \frac{h}{2}g''(x_0) + o(h)$, which would imply that the error term is $\frac{h}{2}g''(x_0) + o(h)$, which indeed is bounded by $C|h|$ for a certain constant $C > 0$.

As $\frac{df}{dx}(x_0, y_0, z_0) = g'(x_0)$, this method approximates the components of the gradient with the desired accuracy.

b)

We use the same method from the previous section. We have to assume that our function is C^3 , and thus we would have that $g(x) = g(x_0) + (x - x_0)g'(x_0) + \frac{1}{2}(x - x_0)^2g''(x_0) + \frac{1}{6}(x - x_0)^3g'''(x_0) + o((x - x_0)^3)$. The central difference scheme asserts that $g'(x_0) \approx \frac{1}{2h}(g(x_0 + h) - g(x_0 - h)) = \frac{1}{2h}(g(x_0) + hg'(x_0) + \frac{1}{2}h^2g''(x_0) + \frac{1}{6}h^3g'''(x_0) - g(x_0) + hg'(x_0) - \frac{1}{2}h^2g''(x_0) + \frac{1}{6}h^3g'''(x_0) + o(h^3)) = g'(x_0) + \frac{1}{6}h^2g'''(x_0) + o(h^2)$, and thus the error term would be bounded by $C|h|^2$ for a certain constant C and sufficiently small values of h .